

Optimal Execution under Market Impact

Market State Representation and Execution Environment

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Abstract

This section develops a realistic market execution environment for reinforcement learning-based optimal execution under market impact.

Using intraday SPY data sampled at 5-minute intervals, the framework integrates market microstructure features, inventory-aware state representations, participation-based temporary market impact, and benchmark execution policies including TWAP and VWAP-like strategies.

The environment was designed to simulate realistic execution dynamics under liquidity constraints and sequential decision-making conditions. Benchmark simulations exhibited financially coherent behavior consistent with classical execution theory and empirical market microstructure observations.

In particular, liquidity-aware execution policies generated smoother participation trajectories and more stable inventory liquidation patterns, highlighting the importance of volume dynamics in execution quality.

The resulting framework provides the foundation for training a Deep Q-Network execution agent capable of learning adaptive execution strategies directly from market state dynamics.

1 Market State Representation and Execution Environment

1.1 Intraday Market Data

To simulate realistic execution dynamics, intraday SPY data sampled at 5-minute intervals was obtained through the Alpha Vantage API. Only regular trading hours were retained, from 09:30 to 15:55 EST, in order to avoid overnight discontinuities and extended-hours liquidity distortions.

The dataset was cleaned by sorting timestamps chronologically, removing duplicated timestamps, and filtering regular market sessions only. The resulting dataset represents the observable intraday market trajectory used by the execution simulator.

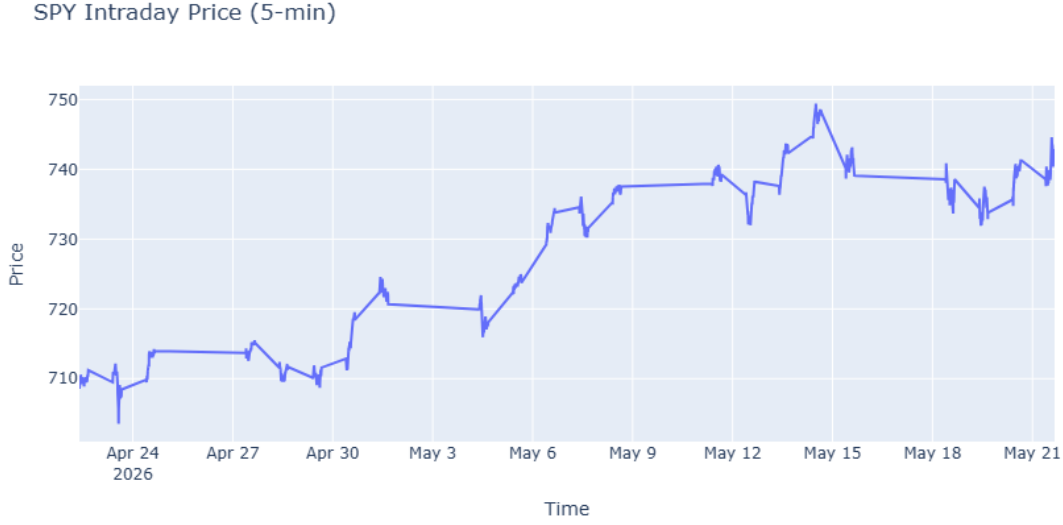


Figure 1: SPY intraday price sampled at 5-minute intervals.

1.2 Market State Features

The execution agent does not directly observe raw prices alone. Instead, the market is represented through engineered features designed to capture relevant market microstructure dynamics associated with execution quality.

The selected feature set includes returns, rolling volatility, relative volume, momentum, spread proxy, and intraday volume profile. These variables jointly characterize liquidity conditions, short-term volatility, trading frictions, directional pressure, and intraday market activity.

Relative volume and volume profile are particularly important because execution quality strongly depends on available liquidity throughout the trading session.

Intraday Volume Profile

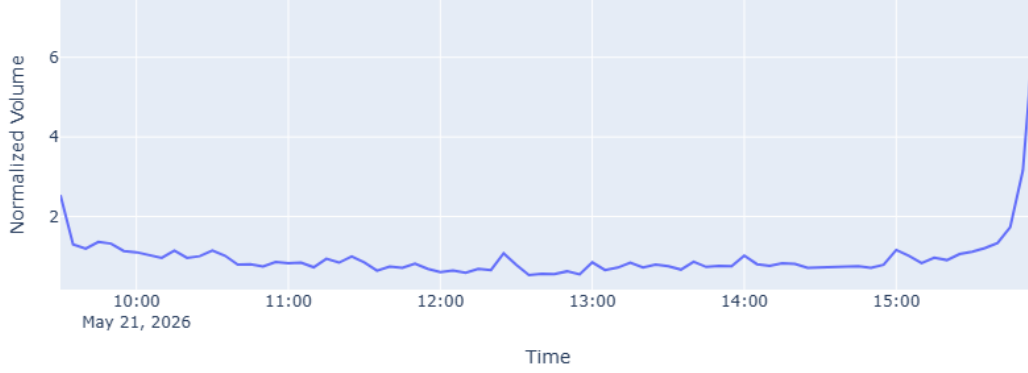


Figure 2: Normalized intraday volume profile exhibiting the classical U-shaped liquidity curve commonly observed in equity markets.

1.3 State Representation

At each timestep t , the environment constructs a state vector containing both execution-related variables and market microstructure information.

$$s_t = \left[\frac{x_t}{Q}, \frac{T-t}{T}, f_t \right]$$

where x_t denotes remaining inventory, Q denotes total parent order size, T denotes the execution horizon, and f_t represents market state features.

The complete state vector is:

$$s_t = [\text{inventory ratio}, \text{time remaining}, \text{relative volume}, \text{rolling volatility}, \text{spread proxy}, \text{momentum}]$$

This representation allows the agent to jointly observe inventory pressure, remaining execution horizon, liquidity conditions, and short-term market dynamics.

1.4 Execution Environment

Execution is modeled as a sequential decision-making process where the agent selects an execution participation rate at every timestep.

The action space consists of discrete execution fractions:

$$a_t \in [0, 0.05]$$

where each action corresponds to a percentage of the total parent order executed during the

current interval.

Executed shares are computed as:

$$q_t = a_t Q$$

subject to:

$$q_t \leq x_t.$$

The environment incorporates a temporary market impact model based on participation rate:

$$P_t^{exec} = P_t + \eta \left(\frac{q_t}{V_t} \right)$$

where P_t is the observed market price, V_t denotes market volume, and η controls temporary impact intensity.

Execution cost is measured through an implementation-shortfall-style formulation:

$$C_t = q_t(P_t^{exec} - P_0)$$

where P_0 denotes the arrival price.

The reward function penalizes execution cost, excessive participation, and remaining inventory risk. This design encourages balanced execution behavior under liquidity constraints.

1.5 Benchmark Policies

Before training a reinforcement learning agent, several benchmark execution policies were implemented to validate the environment dynamics.

1.5.1 Random Policy

The Random policy selects execution actions uniformly from the discrete action space without considering market conditions or inventory dynamics. This policy serves as a naive execution baseline.

1.5.2 TWAP Policy

The TWAP policy distributes inventory approximately uniformly across the execution horizon. The target execution size at timestep t is defined as:

$$q_t^{TWAP} = \frac{x_t}{T - t}$$

This produces an approximately linear inventory decay profile.

1.5.3 VWAP-like Policy

The VWAP-like policy adjusts execution intensity according to observed market volume. The target execution size is proportional to expected remaining liquidity:

$$q_t^{VWAP} = x_t \left(\frac{V_t}{\sum_{u=t}^T V_u} \right)$$

This policy concentrates execution during high-liquidity periods and reduces participation during illiquid intervals.

1.6 Benchmark Policy Comparison

Table 1 summarizes the execution performance across benchmark policies under the same market environment and execution constraints.

Table 1: Benchmark execution policy comparison.

Policy	Cum. Cost	Shares Traded	Final Inventory	Avg Participation	Max Participation	Steps
Random	32766.39	100000	0	0.0078	0.0238	40
TWAP	192481.60	100000	0	0.0035	0.0080	78
VWAP-like	226706.29	100000	0	0.0026	0.0028	78

The Random policy exhibits the highest execution aggressiveness, liquidating inventory significantly faster than the benchmark schedules and generating larger participation spikes.

TWAP produces stable inventory liquidation with approximately constant participation across the execution horizon.

VWAP-like execution maintains the lowest average participation rate, indicating that execution is distributed more efficiently across high-liquidity intervals.

Although execution costs vary across market conditions, the benchmark comparison suggests that liquidity-aware execution policies produce smoother participation dynamics and more stable execution behavior.

1.7 Microstructure Validation

The benchmark simulations exhibit financially coherent execution behavior across policies. The Random policy produces unstable execution trajectories and irregular inventory liquidation patterns. TWAP generates an approximately linear inventory decay, consistent with constant participation over time.

VWAP-like execution adapts dynamically to liquidity conditions, concentrating execution during periods of elevated market volume and maintaining lower participation rates on average. The observed intraday liquidity structure exhibits the classical U-shaped volume curve commonly observed in equity markets, supporting the validity of the simulated market environment.

These results suggest that the environment captures meaningful market microstructure dynamics and provides a suitable framework for training a reinforcement learning execution agent.

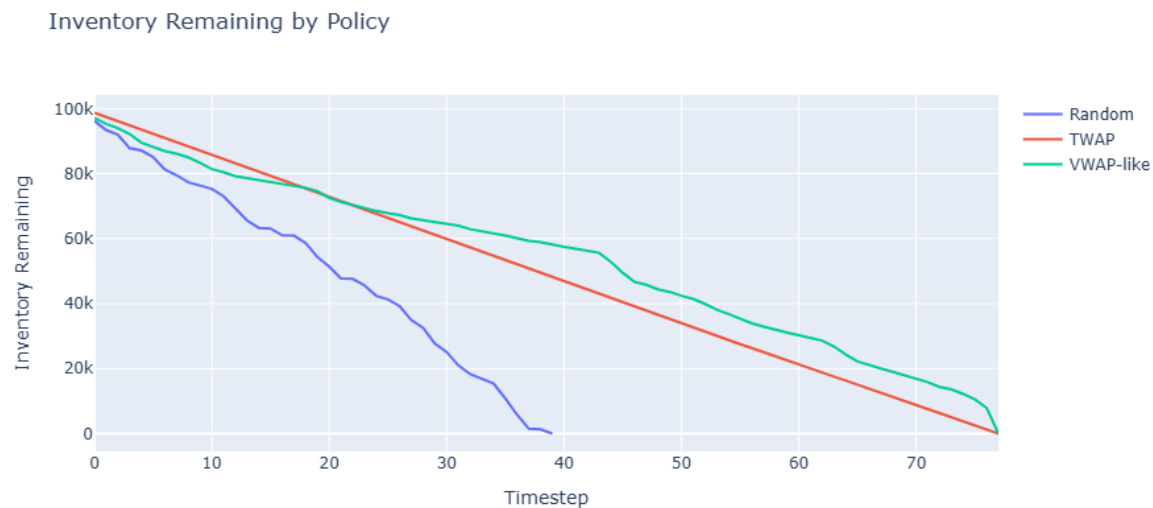


Figure 3: Inventory trajectories across benchmark execution policies.

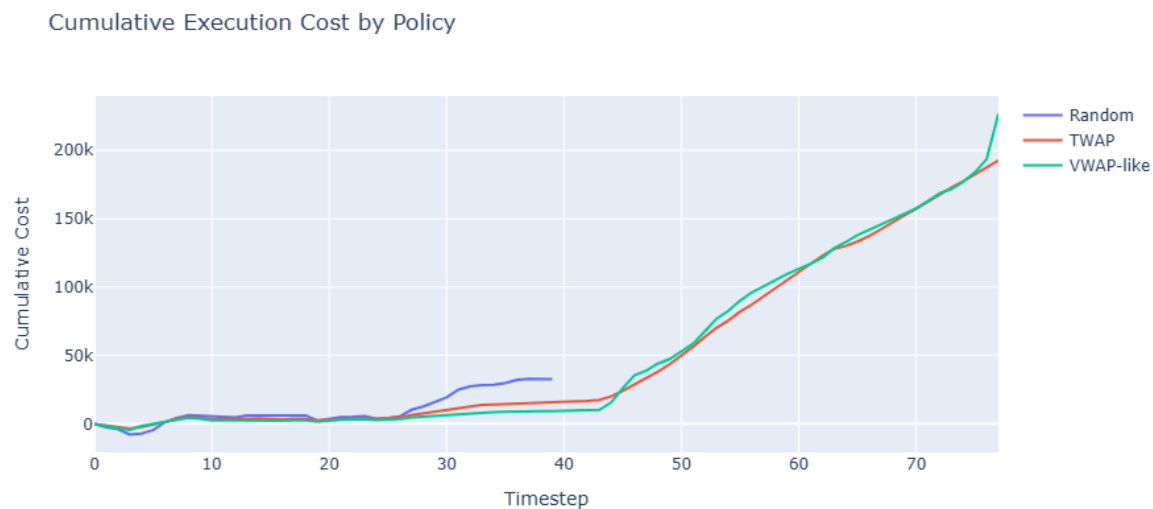


Figure 4: Cumulative execution cost across benchmark policies.

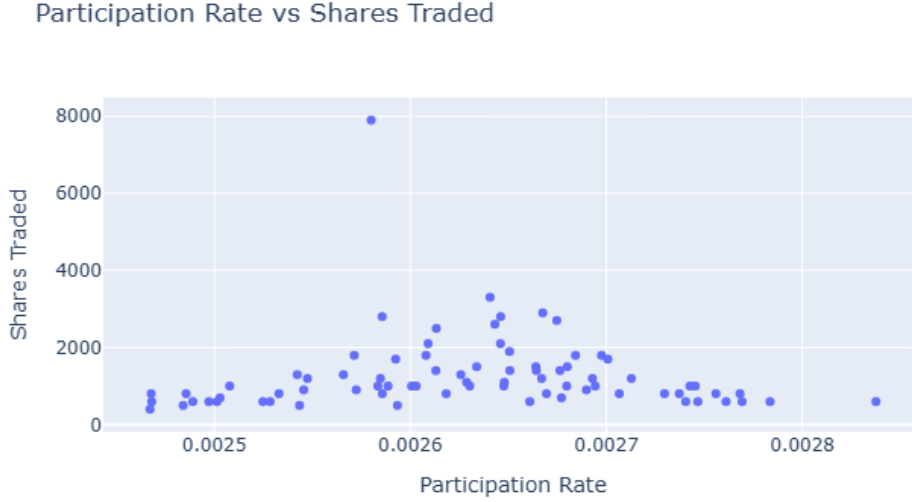


Figure 5: Relationship between participation rate and executed shares under the VWAP-like benchmark policy.

The environment developed in this section provides the foundation for the next stage of the project: training a Deep Q-Network execution agent capable of learning adaptive execution strategies directly from market state dynamics.

2 Key Insights

Several important observations emerged from the execution simulations.

First, execution quality appears to depend strongly on liquidity conditions rather than purely on price dynamics. Volume-aware execution policies consistently produce smoother participation trajectories and more stable inventory liquidation patterns.

Second, the action-space granularity plays a critical role in execution realism. Large execution fractions generated unstable liquidation dynamics, while smaller participation increments produced financially coherent execution schedules.

Third, the benchmark simulations demonstrate that realistic execution modeling requires both inventory-aware and liquidity-aware dynamics. TWAP generates stable but liquidity-insensitive execution, whereas VWAP-like execution adapts naturally to intraday market activity.

Finally, the environment successfully captures several important market microstructure properties, including:

- intraday liquidity variation,
- participation-driven impact,
- execution timing effects,

- inventory pressure dynamics.

These observations support the use of the environment as a suitable framework for reinforcement learning-based execution research.

3 Conclusion

This section developed a realistic execution simulation framework for reinforcement learning-based optimal execution.

The framework combines:

- intraday market state features,
- inventory-aware state representations,
- participation-based temporary market impact,
- benchmark execution policies.

The benchmark simulations exhibited financially coherent behavior consistent with classical execution dynamics observed in real markets.

In particular, liquidity-aware execution policies generated smoother participation trajectories and more stable execution patterns than naive execution approaches.

The resulting environment provides a robust foundation for the next stage of the project: training a Deep Q-Network execution agent capable of learning adaptive execution policies directly from market state dynamics.

References

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